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$$f(x) = x \ln(x) \text{ met } x > 0 \rightarrow 0, -1, 1$$

$$f'(x) = \ln(x) + 1 \rightarrow \frac{1}{e}$$

$$f''(x) = \frac{1}{x}$$

x	0			$\frac{1}{e}$	1		
$f'(x)$	/	-	-	0	+	+	+
$f(x)$	/	$\searrow$		min	$\nearrow$	0	

$$f(x) = x \ln(-x) \text{ met } x < 0$$

$$f'(x) = \ln(-x) + 1 \rightarrow -\frac{1}{e}$$

x	-1			$-\frac{1}{e}$	0			
$f'(x)$	$+$	$+$	$+$	0	$-$	$-$	$-$	$/$
$f(x)$		$\nearrow$		$\max$	$\searrow$			$/$

References